

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Correctly identifies the two end points 90 and 270 Condone $\frac{\pi}{2}$ and $\frac{3\pi}{2}$	AO1.2	M1	$90^\circ < x < 270^\circ$
Correctly uses inequalities May state (90, 270) – must be round brackets Again, condone $\frac{\pi}{2}$ and $\frac{3\pi}{2}$	AO2.5	A1	
Total 2 marks			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Compares with $R \cos(\theta \pm \alpha)$ or $R \sin(\theta \pm \alpha)$	AO3.1a	M1	$R \cos(\theta - \alpha)$ $\equiv R \cos \alpha \cos \theta + R \sin \theta \sin \alpha$ $\therefore R \cos \alpha = 3$ and $R \sin \alpha = 3$ $R = \sqrt{18}$ $\alpha = \frac{\pi}{4}$ $\therefore 3 \cos \theta + 3 \sin \theta \equiv \sqrt{18} \cos \left(\theta - \frac{\pi}{4} \right)$ Which is a stretch in the y-direction scale factor $\sqrt{18}$ $\left(\frac{\pi}{4} \right)$ and a translation $\left(0 \right)$
	Identifies version which will allow them to solve the problem	AO3.1a	A1	
	Obtains correct R	AO1.1b	A1	
	Obtains correct α	AO1.1b	A1	
	Interprets 'their' equation to identify first transformation	AO3.2a	E1	
	Interprets 'their' equation to identify second transformation	AO3.2a	E1	
(b)	Constructs a rigorous mathematical argument, to find either the least or greatest value Only award if they have a completely correct solution, which is clear,	AO2.1	R1	$4 + (3 \cos \theta + 3 \sin \theta)^2$ $4 + \left(\sqrt{18} \cos \left(\theta + \frac{\pi}{4} \right) \right)^2$ Least value occurs when

easy to follow and contains no slips (no FT for this mark)			$\cos^2\left(\theta + \frac{\pi}{4}\right) = 0$ $\therefore \text{least value} = 4$
Deduces the least value Using 'their' values of R and α	AO2.2a	A1F	Greatest value occurs when $\cos^2\left(\theta + \frac{\pi}{4}\right) = 1$
Deduces the greatest value Using 'their' values of R and α	AO2.2a	A1F	Greatest value = $4 + 18$ = 22
Total 9 marks			

Q3.

(a) $x + 30^\circ = 79^\circ$, $x + 30^\circ = 180^\circ + 79^\circ$

$x = 49^\circ$

49 as the only solution in the interval $0^\circ \leq x \leq 90^\circ$

B1

$x = 229^\circ$

AWRT 229. Not given if any other soln. in the interval $90^\circ \leq x \leq 360^\circ$.

Ignore anything outside $0^\circ \leq x \leq 360^\circ$

B1

2

(b) Translation;

*Accept 'translat...' as equivalent.
[T or Tr is NOT sufficient]*

B1

$$\begin{bmatrix} -30^\circ \\ 0 \end{bmatrix}$$

*OE Accept **full** equivalent to vector in words provided linked to 'translation / move / shift' and **correct** direction. (0 / 2 if >1 transformation).*

B1

2

(c) (i) $5 + \sin^2 \theta = (5 + 3\cos \theta) \cos \theta$
 $\Rightarrow 5 + \sin^2 \theta = 5\cos \theta + 3\cos^2 \theta$
Correct RHS.

B1

$$5 + 1 - \cos^2 \theta = 5\cos \theta + 3\cos^2 \theta$$

$\sin^2 \theta = 1 - \cos^2 \theta$ used to get a quadratic in $\cos$.

M1

$$6 = 5\cos \theta + 4\cos^2 \theta \text{ or } 4\cos^2 \theta + 5\cos \theta - 6 (= 0)$$

ACF with like terms collected.

A1

$$\Rightarrow (4\cos \theta - 3)(\cos \theta + 2) (= 0)$$

Correct quadratic and $(4c \pm 3)(c \pm 2)$ or by formula
OE PI by $\pm$ 'correct' 2 values for $\cos \theta$.

m1

Since $\cos \theta \neq -2$, $\cos \theta = \frac{3}{4}$

CSO AG. Must show that the 'soln' $\cos \theta = -2$ has been considered and rejected

A1

5

(ii) $5 + \sin^2 2x = (5 + 3\cos 2x) \cos 2x \Rightarrow \cos 2x = \frac{3}{4}$

Using (c)(i) to reach $\cos 2x = \frac{3}{4}$ **or** finding at least 3 solutions of $\cos \theta = \frac{3}{4}$ **and** dividing them by 2.

M1

$$2x = 0.722(7...), 2\pi - 0.722(7...), \\ 2\pi + 0.722(7...), 4\pi - 0.722(7...),$$

Valid method to find all four 'positions' of solutions.

m1

$$x = 0.361, 2.78, 3.50, 5.92$$

CAO Must be these four 3sf values but ignore any values outside the interval $0 < x < 2\pi$.

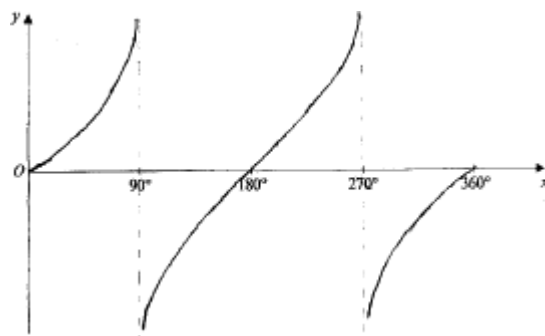
A1

3

[12]

Q4.

(a) (i)



Ignore any part of the graph drawn outside interval $0^\circ \leq x \leq 360^\circ$ in (a)

A 3 branch curve between 0 and 360 meeting the x-axis at or very close to 0, 180, 360 only

B1

A 3 branch curve between 0 and 360 with correct shape tending to infinity at, at least 3, of the 4 relevant ends

B1

Correct graph for $0^\circ \leq x \leq 360^\circ$ with correct intercepts. Asymptotes not explicitly required but graphs should show correct 'tendency' close to 90 and 270.

B1

3

(ii) 135° ; 315°

B2 for both 135 and 315 and no 'extra' in interval $0^\circ \leq x \leq 360^\circ$ (If not B2 then award B1 for either 135 or 315 with or without extras)

B2,1,0

2

(c) (i) $6 \tan \theta \sin \theta = 5 \Rightarrow 6 \frac{\sin \theta}{\cos \theta} \sin \theta = 5$
 $\tan \theta = \frac{\sin \theta}{\cos \theta}$ used

M1

$$6 \frac{\sin^2 \theta}{\cos \theta} = 5 \Rightarrow 6 \frac{1 - \cos^2 \theta}{\cos \theta} = 5$$

$\sin^2 \theta$ replaced by $1 - \cos^2 \theta$ throughout

m1

$$6 - 6\cos^2 \theta = 5\cos \theta \Rightarrow 6\cos^2 \theta + 5\cos \theta - 6 = 0$$

Completion AG Be convinced

A1

(ii) $6 \tan 3x \sin 3x = 5 \Rightarrow 6 \cos^2 3x + 5 \cos 3x - 6 = 0$

Using (b)(i) with $\theta = 3x$ PI by attempting to solve eg for theta then dividing soln(s) by 3

M1

$$(3\cos 3x - 2)(2\cos 3x + 3) (= 0)$$

$$(\cos 3x = 2/3, -3/2)$$

Correct factorisation or correct subst into the quadratic formula PI by two 'correct' roots

m1

$$\cos 3x = \frac{2}{3} = \cos 48.1(89..) [= \cos \alpha]$$

$$3x = \alpha, 360 - \alpha, 360 + \alpha.$$

Dep on M1 only,

$3x = \alpha, 360 - \alpha, 360 + \alpha$ for c's α . from an eqn $\cos 3x = k$ where $-1 < k < 1$ OE PI and no solns from k outside $-1 \leq k \leq 1$

m1

$$x = 16^\circ,$$

B1

$$104^\circ,$$

B1

$$136^\circ$$

B1

AWRT 16, 104, 136. Deduct one mark (from any award of these 3 B marks) if more than three solns given inside the interval $0^\circ \leq x \leq 180^\circ$.

Ignore any solutions outside the interval $0^\circ \leq x \leq 180^\circ$.

NMS Max. is B3/6

Q5.

(a) (i) $h^2 = 2^2 + \sqrt{5}^2 = 9 \Rightarrow h = 3 \Rightarrow \sin \alpha = \frac{2}{3}$

Pythagoras used or all of 2, $\sqrt{5}$, 3 seen correctly on triangle

B1

$$\cos \alpha = \frac{\sqrt{5}}{3}$$

AG $\frac{\sqrt{5}}{3}$ or $\sqrt{\frac{5}{9}}$ or $\frac{5}{3\sqrt{5}}$ seen

B1

2

Alternative:

$$\operatorname{cosec}^2 \alpha = 1 + \cot^2 \alpha = 1 + \frac{5}{4} = \frac{9}{4}$$

$$\operatorname{cosec} \alpha = \frac{3}{2} \quad \sin \alpha = \frac{2}{3}$$

Must be positive

(B1)

$$\sec^2 \alpha = 1 + \tan^2 \alpha = 1 + \frac{4}{5} = \frac{9}{5}$$

$$\sec \alpha = \frac{3}{\sqrt{5}} \quad \cos \alpha = \frac{\sqrt{5}}{3}$$

Must be positive

(B1)

(2)

(ii) $\sin 2\alpha = 2 \sin \alpha \cos \alpha$

Correct formula seen or implied

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M1

$$= \left(2 \times \frac{2}{3} \times \frac{\sqrt{5}}{3} \right) = \frac{4}{9} \sqrt{5}$$

Must see $\frac{\sqrt{5}}{3}$ here or in part (a)(i)

Accept $\frac{4}{3} \sqrt{\frac{5}{9}}$

A1

2

(b) $\cos \beta = \frac{2}{\sqrt{5}}$ or $\sin \alpha = \frac{1}{\sqrt{5}}$

Either correct. Accept $\sqrt{\frac{4}{5}}$, $\frac{\sqrt{5}}{5}$

B1

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

Correct formula seen or implied.

M1

$$= \frac{\sqrt{5}}{3} \times \frac{2}{\sqrt{5}} + \frac{2}{3} \times \frac{1}{\sqrt{5}}$$

All correct

A1

$$= \frac{2}{15} (5 + \sqrt{5})$$

k = 5 with previous A mark awarded

A1

4

[8]

Q6.

(a) (i) $\tan \alpha = \frac{4}{3}$

Fraction required
Allow 1.333 (recurring)

B1

1

(ii) 1, 2, $\sqrt{3}$ seen (from Pythagoras)
or
 $4 = 1 + \cot^2 \beta$
Use cosec²β = 1 + cot² β

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$$\tan \beta = -\frac{1}{\sqrt{3}}$$

SC B1 $\frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$

A1

Special case B1 for $\frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$

2

$$(b) \quad \tan(\alpha + \beta) = \frac{\frac{4}{3} - \frac{1}{\sqrt{3}}}{1 - \frac{4}{3} \left(-\frac{1}{\sqrt{3}} \right)}$$

Use $\tan(\alpha + \beta)$ formula

M1

Remove fractions within fractions

Correct manipulation to form $\frac{a+b\sqrt{3}}{c+d\sqrt{3}}$ $a \ b \ c \ d$ integers

m1

$$= \frac{4\sqrt{3} - 3}{3\sqrt{3} + 4}$$

$m = 4 \quad n = 3$
or any multiple

A1

3

Alternative $\tan(\alpha + \beta)$

$$\frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{\frac{4}{5} \times \left(-\frac{\sqrt{3}}{2} \right) + \frac{3}{5} \times \frac{1}{2}}{\frac{3}{5} \times \left(-\frac{\sqrt{3}}{2} \right) - \frac{4}{5} \times \frac{1}{2}}$$

Use formulae for $\sin(\alpha + \beta)$ and $\cos(\alpha + \beta)$

(M1)

Remove fractions within fractions

Correct manipulation to form $\frac{a+b\sqrt{3}}{c+d\sqrt{3}}$ $a \ b \ c \ d$ integers

(m1)

$$= \frac{-4\sqrt{3} + 3}{-3\sqrt{3} - 4} \quad \left(= \frac{4\sqrt{3} - 3}{3\sqrt{3} + 4} \right)$$

$m = -4 \quad n = -3$
or any multiple

(A1)

(3)

*M1 for substituting candidates values for $\tan \alpha$ and $\tan \beta$ into correct formula.
Completely correct or completely correct ft on $\tan \alpha$, $\tan \beta$.*

Special case answer is $\frac{12+3\sqrt{3}}{9-4\sqrt{3}}$ or $\times \frac{a}{a}$ where a is integer or $\sqrt{3}$ for M1m1A0.

[6]

Q7.

(a) (Area) = $\frac{1}{2}(26)(31.5)\sin \theta$
 $\frac{1}{2}(26)(31.5)\sin (\theta)$ stated or used

M1

$$\frac{1}{2}(26)(31.5) \times \frac{5}{13} = 157.5 \text{ (cm}^2\text{)}$$

OE eg $\frac{315}{2}$

Condone AWRT 157.50

NMS: 157.5 or AWRT 157.50 scores B2

A1

2

(b) $(\cos \theta =) \frac{12}{13}$
 $\frac{12}{13}$ OE exact fraction

B1

1

(c) $\{AC^2 =\}$
 $31.5^2 + 26^2 - 2 \times 31.5 \times 26 \times \cos (\theta) = 992.25 + 676 - 1512$
 RHS of cosine rule

M1

$$= 1668.25 - 1512 = 156.25$$

Correct order of evaluation. Do not award if evaluation leads to or would lead to RHS value being outside interval 120 to 195

m1

$$AC = \sqrt{156.25} = 12.5 \text{ (cm)}$$

12.5 OE with no sight of premature approximation clearly used

A1

3

Alternative

$$\{AC^2 = \} (26 \sin \theta)^2 + (31.5 - 26 \cos \theta)^2$$

(M1)

$$= 10^2 + 7.5^2$$

(m1)

$$AC = \sqrt{156.25} = 12.5 \text{ (cm)}$$

(A1)

(3)

[6]

Q8.

(a) (i) $\sin 2\theta = 2 \sin \theta \cos \theta$

B1

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

OE condone use of x etc, but variable must be consistent

B1

2

(ii) $\sin \theta = \frac{4}{5} \Rightarrow \sin 2\theta = 2 \times \frac{4}{5} \times \frac{3}{5} = \frac{24}{25}$

AG

Use of 106.26° B0

B1

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or

$$2 \times \sin \left(\cos^{-1} \frac{3}{5} \right) \times \frac{3}{5}$$

$$\cos 2\theta = \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}$$

$$-0.28$$

B1

2

(b) (i) $\frac{dx}{d\theta} = 6 \cos 2\theta \quad , \quad \frac{dy}{d\theta} = -8 \sin 2\theta$

Attempt both derivatives. ie $p \cos 2\theta$

M1

Both correct.

$$q \sin 2\theta$$

A1

$$\frac{dy}{dx} = -\frac{4}{3} \frac{\sin 2\theta}{\cos 2\theta} \quad \text{ISW}$$

CSO OE

A1

3

(ii) $P\left(\frac{72}{25}, -\frac{28}{25}\right)$
 $(2.88, -1.12)$

B1F

$$\text{Gradient} = -\frac{4}{3} \times -\frac{24}{7}$$

$\frac{q \sin 2\theta}{p \cos 2\theta}$ or $\frac{p \cos 2\theta}{q \sin 2\theta}$
 Their
 must be working with rational numbers

M1

Tangent $y + \frac{28}{25} = \frac{32}{7} \left(x - \frac{72}{25}\right)$ ISW

Any correct form.

$$7y = 32x - 100$$

Fractions in simplest form

Equation required

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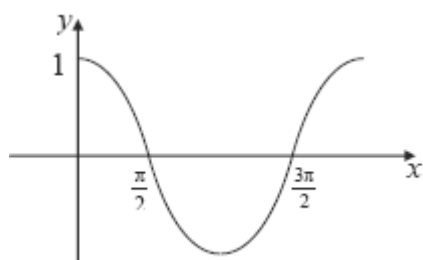
A1

3

[10]

Q9.

(a)



Correct shape meeting positive y-axis

and only one oscillation within interval
0 to 2π

B1

The three correct intercepts stated;
Accept 1.57 for $\pi/2$ and 4.71 for $3\pi/2$
but must be evidence of radian vals.
not just degrees

B1

Ignore any parts of the graph clearly
indicated as outside the given interval

2

(b) (i) $1 - \cos^2 \theta = \cos \theta(2 - \cos \theta)$
 $\cos^2 \theta + \sin^2 \theta = 1$ used

M1

$$1 = 2 \cos \theta \Rightarrow \cos \theta = \frac{1}{2}$$

CSO AG Completion

A1

2

(ii) $\sin^2 2x = \cos 2x(2 - \cos 2x) \Rightarrow \cos 2x = \frac{1}{2}$
Uses (b)(i)

M1

$$\{2x =\} \cos^{-1}\left(\frac{1}{2}\right) = 1.04(7..)$$

PI Accept 1.05, $\frac{\pi}{3}$; Condone 60°

m1

$$x = 0.524, 2.62$$

$$x = 0.523(59..), 2.61(7..)$$

Condone > 3sf; Condone $x = 0.525, 2.62$
Accept truncated '3sf' vals $x = 0.523, 2.61$
Deduct 1 mark for each extra (> 2 solns) in
the given interval from A marks to a min
of A0. Ignore any solns outside the given
interval 0 to π .
Accept, as equivalent, the exact answers

$x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$ when seen and apply
 ISW if 'errors' converting these to decimals.

If not A2 then A1 if

- one soln correct.
- 30° , 150° ie solns left in degrees
- AWRT 0.52, 2.6 ie correct vals to only 2sf.

Must see an indication that (b)(i) has been
 used otherwise 0/4 so just stating the two
 correct answers with nothing else scores 0/4.

A2,1,0

4

[8]

Q10.

(a) $\{x = \} \sin^{-1}(0.8) = 0.927(29\dots) \{ = \beta \}$
 $\sin^{-1}(0.8)$ PI

M1

$\{x = \} \pi - \beta$

m1

$x = 0.927(29\dots), 2.21(42\dots)$

Both

Ignore values outside interval $0 - 2\pi$ but
 A0 if 'extra' values inside the given interval

A1

3

(b) (i) $\left(\frac{3\pi}{2}, -1\right)$

$B1$ if one coordinate correct or $\left(-1, \frac{3\pi}{2}\right)$

B2,1

2

(ii) $\pi - \alpha$

B1

1

(iii) $RS = (2\pi - \alpha) - (\pi + \alpha)$

OE eg $RS = PQ = (\pi - \alpha) - \alpha$

M1

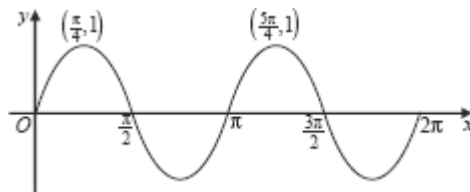
$$= \pi - 2\alpha$$

Must be simplified

A1

2

(c)



*Sine curve with positive gradient at O
with at least 3 stationary points between 0
and 2π*

B1

*Correct shaped curve with 2 max and 2
min between 0 and 2π*

B1

*All 5 correct points of intersection with
 x -axis with $\frac{\pi}{2}$, π and $\frac{3\pi}{2}$ clearly shown*

B1

Maximum points $\left(\frac{\pi}{4}, 1\right)$ and $\left(\frac{5\pi}{4}, 1\right)$
stated or clearly shown on the sketch

B1 for either:

*1 as the y -coordinate of max pt(s) or:
two max pts between 0 and 2π with
correct x -coordinates*

B2,1

5

[13]

Q11.

(a) (i) $\cos \alpha = \frac{3}{5}$
ACF

B1

1

(ii) $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

M1

$$= \frac{3}{5} \cos \beta + \frac{4}{5} \sin \beta$$

ACF

A1

2

(iii) $\sin \beta = \frac{12}{13}$

B1

$$\cos(\alpha - \beta) = \frac{63}{65}$$

$$\frac{63}{65} \quad \text{NMS B1 B1}$$

B1

2

(b) (i) $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

M1

$$2 \tan x = 1 - \tan^2 x$$

$$\tan^2 x + 2 \tan x - 1 = 0$$

CSO AG

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A1

2

(ii) $\tan x = \frac{-2 \pm \sqrt{4 + 4}}{2}$

*must solve quadratic equation by formula or by
completing the square
condone one slip*

M1

$$= -1 \pm \sqrt{2}$$

$$\pm \sqrt{2} \text{ required}$$

A1

$$2x = 45^\circ \Rightarrow x = 22\frac{1}{2}^\circ \text{ is acute}$$

$$\Rightarrow \tan 22\frac{1}{2}^\circ = \sqrt{2} - 1$$

explain selection of positive root

E1

3

[10]

Q12.

- (a) $\{x = \} \cos^{-1}(0.3) = 1.266\dots \{ = \beta \}$
 $\cos^{-1}(0.3)$ PI by eg 72° or 73°

M1

$$\{x = \} 2\pi - \beta$$

Condone degrees or mix.

m1

$$x = 1.27, 5.02$$

Accept 1.26 to 1.27 with 5.01 to 5.02 inclusive

A1

3

- (b) (i) $m(\pi, -1)$

B1 for each coordinate

B1;B1

2

- (ii) $\{x_Q = \} = 2\pi - \alpha$

OE (unsimplified)

B1

1

- (c) Stretch (I) in x -direction (II) scale

Need(I) & one of (II),(III)

M1

factor $\frac{1}{2}$ (III)

A1

2

(d) $\cos 2x = \cos \frac{4\pi}{5} \Rightarrow 2x = \frac{4\pi}{5}$

OE. (From correct work)

B1

$\Rightarrow x = \frac{2\pi}{5} (= \alpha)$

Condone decimals/degrees

$x = \pi - \alpha$; OE

OE eg $2x = 2\pi - \frac{4\pi}{5}$

Correct quadrant;
condone degrees/decimals/mix

M1

$x = \pi + \alpha$; $x = 2\pi - \alpha$; OE

Need both (OE for $2x =$) with no extras
(quadrants) within the given interval.
Condone degrees/decimals/mix

m1

$x = \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{7\pi}{5}, \frac{8\pi}{5}$

Need all 4 solutions for x but condone
unsimplified provided in terms of π
Ignore extra values outside the given interval.

A1

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4

[12]

Q13.

(a) $A(0^\circ, 1)$

Condone radians

B1

$B(45^\circ, 0)$

Condone (0.785,0) or better.

B1

$C(270^\circ, -1)$

B1 for 270; B1 for -1

B1, B1

4

- (b) Stretch (I) in x -direction (II) with a scale factor $\frac{1}{2}$ (III)
More than one transformation is M0
*M1 for (I) **and** either (II) or (III)*

M1A1

2

- (c) $\cos^{-1}0.37 = "68.284..." (= \alpha)$
Cos⁻¹0.37 (PI eg by 68.3 or 1.19)

M1

$$x = \frac{\alpha}{2} = 34.1(42.)^\circ$$

Condone 34.2°, 34° or 0.596 rads

A1

$$x = 180 - \frac{\alpha}{2}$$

OE eg $2x = 360 - \alpha$

m1

$$x = 180 + \frac{\alpha}{2} \quad \text{and} \quad x = 180 + 180 - \frac{\alpha}{2}$$

*OE Need both (OE for $2x =$)
 with no extras (quadrants)
 within the given interval*

m1

$$2x = 38.284...; 291.175...;$$

$$428.284...; 651.715...$$

$$x = (34.1^\circ;)$$

$$145.9^\circ; 214.1^\circ; 325.9^\circ$$

*Dep. on **all** three method marks. Must be in degrees*

A1

5

[11]